

AICTE – EDUSKILLS VIRTUAL INTERNSHIP

Python Full Stack Development & Project

An 8-Week Professional Internship Presentation Report

📅 June 2026 – August 2026

Candidate & Program Overview

Student Name	Girik Goswami
Roll Number	2410031637
Academic Program	B.Tech (Computer Science & Engineering)
University	IILM University, Greater Noida, U.P.
Internship Domain	Python Full Stack Development With Project
Host Organization	EduSkills Academy (Supported by AICTE National Internship Portal)

Problem Statement & Industry Context

The Industry Challenge

Developers face significant challenges in efficiently updating or deleting records in memory without experiencing severe performance degradation.

Furthermore, real-world systems require robust error-handling mechanisms to gracefully manage malformed, incomplete, or inconsistent JSON inputs during complex record manipulation tasks.

The Solution Scope

The JSON Database Simulator directly addresses these constraints by providing a lightweight, highly efficient in-memory NoSQL database system.

It processes various user commands through a text-based interface, supporting core CRUD operations, sophisticated data filtering, and generating system snapshots while returning all outputs in a standardized JSON format.

Core Project Objectives

Design & Build



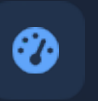
Design and implement a fully functional, highly responsive in-memory NoSQL database simulator capable of robust data handling.

Modern Tools



Utilize modern programming tools, efficient algorithms, and Pythonic techniques (like list comprehensions) to manipulate structured data.

Performance



Optimize underlying system performance and code quality to ensure rapid data retrieval, filtering, and modification operations.

Technologies & Toolchains

Mastering an enterprise-grade stack to construct robust full-stack applications:

- **Core Languages:** Python, JSON for data serialization
- **Backend & APIs:** FastAPI framework, Pydantic for validation, SQLAlchemy ORM
- **Frontend:** React, React Hooks, React Router, Redux/Zustand
- **Data Structures:** Python Dictionaries, Lists
- **Testing & Deployment:** Docker containerization, Cloud platforms



System Architecture

The JSON Database Simulator relies on an in-memory data structure built upon Python dictionaries, where distinct keys represent specific named databases.

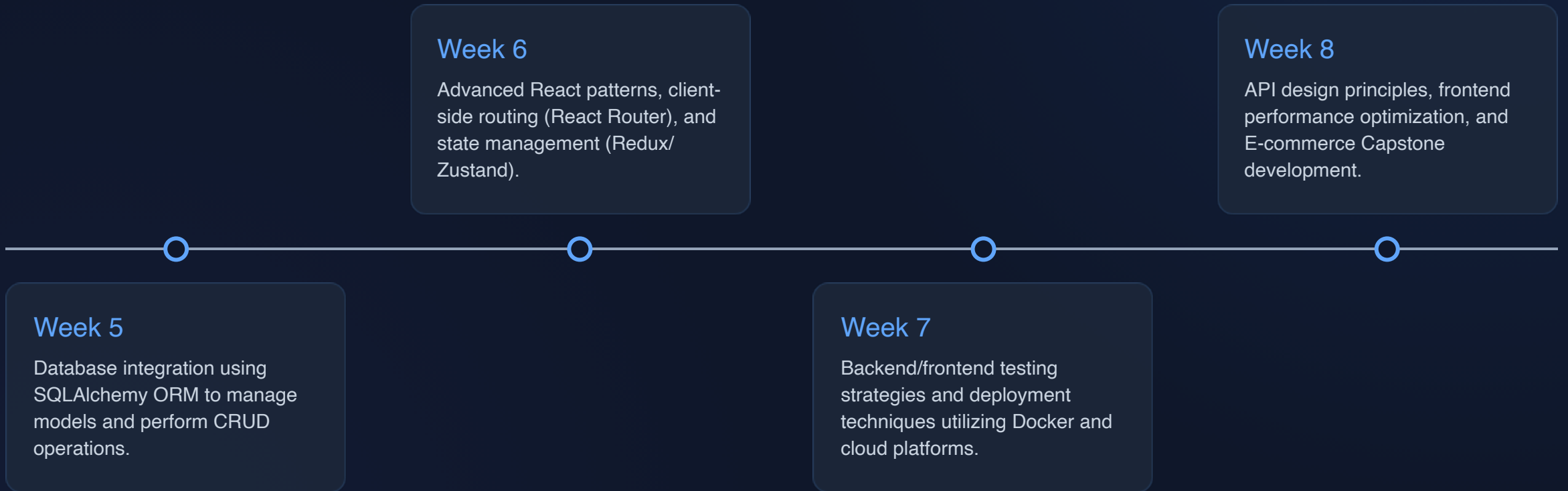
Database records are stored internally as lists of JSON objects. Operations execute via a text-based interface that parses inputs, utilizing list comprehensions and lambda functions for efficient querying, filtering, and snapshot generation.



Curriculum Timeline (Weeks 1 to 4)



Curriculum Timeline (Weeks 5 to 8)



Key Projects & Demonstrated Skills



JSON Database Simulator Developed an in-memory NoSQL database utilizing Python dictionaries and JSON parsing for advanced data persistence, manipulation, and programmatic filtering.



E-commerce Capstone Project Built and deployed a comprehensive full-stack platform featuring seamless user authentication, dynamic product catalogs, and secure relational database integration.



Core Competencies Acquired Mastered API Design Principles, robust Frontend Optimization techniques, Data Serialization, complex State Management, and end-to-end Full Stack Integration.

Certification & Official Verification

This credential validates practical skills in modern Python full stack engineering methodologies.

- **Issuing Organization:** EduSkills Academy
- **Internship Domain:** Python Full Stack Development With Project
- **Duration:** 8 Weeks (June 2026 - August 2026)
- **Certificate ID:** 2026-33983D53FC



Thank You!

Girik Goswami

Roll No: 2410031637

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